



I am a researcher and developer with hands-on experience building deep learning models based on diffusion models, GANs, and VLM-based systems on multi-GPU platforms (CUDA). My work has led to multiple publications, including a new model submitted to CVPR 2026. I bring strong skills in PyTorch, computer vision, model optimization, and AI agents.

Content

- (PT) +351-912292634
- Farhadsh1992@gmail.com
- Citizen of Portugal and Iran.
- Portugal
- www.linkedin.com/in/farhadsh1992
- www.github.com/farhadsh1992/

Language

English (Fluent)

Farsi (Native)

Portuguese (basic)

Programming

Python, C++, Java (Basic)

AI/ML: **PyTorch**, **TensorFlow**, ONNX,

PyTorch Lightning, **huggingface**,

DeepSpeed, langchain, langsmith,

langgraph

Multi-GPU, CUDA,

OpenCV, **Pillow**, **Scikit-learn**

SQL, **MySQL**, **Spark**, **pandas**, **ChromaDB**

Project

- VISUAL-ID – Unique Visual Identities in Graphics, Images and Faces ([Link](#))
- TruIM – Trust Image ([Link](#))
- FACING ([link](#))
- link:
<https://visteam.isr.uc.pt/team/farhad-shadmand-2/>

Entertainment

- Dance (Afro-Latin, Forro)
- Swimming
- Gaming
- Movie

FARHAD SHADMAND

AI Researcher and Developer



Experience

AI Researcher Internship

Aug-
2025
-
Nov
2025

Adobe – Paris, France.

- Developed robust watermarking models using geometric symmetry-aware convolutional networks combined with a VLM-based generative noise-simulation pipeline to accurately replicate print-and-scan degradation effects. This approach significantly enhanced watermark resilience against real-world distortions introduced during printing, scanning, and image recapture.

- Managers:** Dr. Shruti Agarwal (shragarw@adobe.com).

AI Researcher

2019
-
2026

VisTeam – Instituto de Sistemas e Robótica – Coimbra, (ISR)

- Developed deep learning models for steganography, watermarking, object detection, face detection, and face verification. Contributed to the VISUAL-ID, TruIM, and FACING projects, resulting in multiple publications and practical solutions for identity-document security.



Education

PhD. In Electrical Engineering and Intelligent Systems

2021
-
2026

University of Coimbra, Portugal

- My PhD focused on developing printable steganography and **watermarking models** for identity-document security, combining encoder-decoder architectures, geometric-symmetric convolutions, and advanced noise-simulation pipelines. I designed and optimized several deep-learning frameworks for reliable message embedding under real print-and-capture distortions. The work resulted in multiple publications, including IEEE Access, CVPR, and WACV.
- Supervisors:** Prof. Nuno Gonçalves (nunogon@deec.uc.pt) and Prof. Luiz Schirmer (Luiz.schirmer@ufsm.br)

M.Sc. in Complex system, Physics

2015
-
2017

Isfahan University of Technology, Iran

- I specialize in predicting and modeling stock market behavior, utilizing advanced data analysis and machine learning techniques to forecast trends and inform investment strategies.
- Supervisor:** Prof. Farhad Shahbazi (shahbazi@iut.ac.ir)

Bachelor in Physics

2011
-
2015

Isfahan University of Technology, Iran

Among the top 10% of graduates.



Innovation patent

- Inventors: Nuno Gonçalves and Farhad Shadmand, Applicants: Imprensa Nacional Casa da Moeda and University of Coimbra. Encoding, decoding and integrity validation systems for a security document with a steganography-encoded image and methods, computer programs and associated computer-readable data carrier. In PT117136A (Portugal), EP4064095A1 (European Patent Office). **Granted in Oct. 2025.**



Papers References

- StampOne: Addressing Frequency Balance in Printer-proof Steganography – CVPR – 2024.**
- CodeFace: A Deep Learning Printer-Proof Steganography for Face Portraits. IEEE Access 9 (2021)**
- StylePuncher: Encoding a Hidden QR Code into Images, ICPRAM (2025),**
- DocSafe: Towards Practical Print-Proof Image steganography via Frequency Decomposition and Covariance Alignment”, IEEE Access (2026.3680290), 2026**
- RiemStega: Covariance-based loss for print-proof transmission of data in images, WACV (2025).
- Young Labeled Faces in the Wild (YLFW): A Dataset for Children Faces Recognition – IEEE 18th International Conference on Automatic Face and Gesture Recognition (FG) – (2024)
- Mordeephy: Face morphing detection via fused classification-12th International Conference on Pattern Recognition Application and Methods – (2022)
- Towards facial biometrics for id document validation in mobile devices. – Applied Sciences 11.13 (2021)